CINNAMONSRI: A SMART IOT-INTEGRATED SYSTEM FOR ENHANCING CEYLON CINNAMON PRODUCTION AND EXPORT EFFICIENCY

25-26J-319

Project Proposal Report

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AI-POWERED QUALITY & PRICE PREDICTION

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Declaration

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously publish or written by another person expect where the acknowledgement is made in the text.

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Abstract

Sri Lanka's Ceylon cinnamon industry faces significant challenges in quality grading and pricing due to reliance on manual processes and middlemen, leading to inconsistent yields and reduced profitability for smallholder farmers. This proposal introduces the AI/ML-Based Cinnamon Quality and Price Prediction Tool, a component of the CinnamonSri system, enabling farmers to input parameters such as temperature, moisture content, bark thickness, pre- and post-drying weights, and visual attributes via an intuitive mobile or web application. Utilizing Random Forest and XGBoost models, trained on a 1,000–5,000-point dataset from collection centers, agricultural institutions, farmer surveys, and synthetic data adhering to ISO 6539:2014 standards, the tool classifies quality grades (Premium, Grade A, Grade B) and predicts prices (LKR/kg). Data preprocessing includes normalization, 80/20 training/testing split, cross-validation, and evaluation with Mean Squared Error (<10% variance) and F1-score (>85% expected). Deployment options include local devices with Sinhala/Tamil interfaces or cloud-basedAPIs (e.g., AWS).

Anticipated results include accurate predictions enhancing farmer decision-making, potentially increasing profits by 20–30% by reducing middlemen dependency. Conclusions highlight the tool's novelty as a low-cost, IoT-independent AI solution for localized agriculture, with recommendations for regular model updates with market data and field testing for broader adoption. This IT-driven approach bridges technological gaps, promoting sustainable cinnamon production.

Keywords: Machine Learning, Cinnamon Quality Grading, Price Prediction, Agricultural AI, Farmer Decision Support

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List of abbreviations

AI	Artificial Intelligence
ML	Machine Learning
LKR	Sri Lankan Rupee
API	Application Programming Interface
AWS	Amazon Web Services
UI	User Interface
IDE	Integrated Development Environment
VCS	Version Control System
VS Code	Visual Studio Code

1. INTRODUCTION

Sri Lanka, known as the "Spice Island," is a global leader in producing Ceylon cinnamon (Cinnamomum verum), a premium spice valued for its unique aroma, taste, and low coumarin content. The cinnamon industry supports over 400,000 livelihoods and generates approximately USD 200 million annually in export revenue [5]. Quality grading and pricing of cinnamon are critical to maintaining its market value, yet smallholder farmers, who form the backbone of this industry, face significant challenges due to reliance on traditional methods [2]. This research proposes a low-cost, AI/ML-based quality and price prediction tool to provide automated grading and price estimation with user-friendly interfaces, tailored for smallholder cinnamon farmers in Sri Lanka.

1.1 Background and Literature Survey

1.1.1 Background

Ceylon cinnamon's high demand in international markets stems from its superior quality, but achieving consistent grading and pricing requires precise post-harvest assessment, particularly based on attributes like moisture content, weight differences, and bark characteristics [4]. Smallholder farmers predominantly use manual grading due to its low cost and simplicity. However, this method is subjective, leading to inconsistent classifications, exploitation by middlemen, and reduced economic returns, which lower the export price [3], [4]. Unpredictable market trends and lack of real-time data exacerbate these issues, causing financial losses and reduced competitiveness [2]. To address these challenges, researchers have explored AI and machine learning technologies, but their cost and complexity limit adoption by small-scale farmers. The emergence of artificial intelligence (AI) and machine learning (ML) offers a promising avenue for developing affordable, predictive tools that can enhance decision-making and accessibility for rural farmers.

1.1.2 Literature Survey

The literature on cinnamon quality assessment, pricing, and AI/ML applications in agriculture provides critical insights into existing technologies and highlights gaps that this research aims to address. Traditional manual grading, while cost-effective, results in inconsistent quality due to human error and subjectivity. Munasinghe [3] found that manual classification of Ceylon cinnamon reduces accuracy in distinguishing grades, impacting its market value. Perera [4] further noted that manual methods lead to uneven quality evaluation, causing price discrepancies, particularly in volatile markets.

Advanced quality and pricing technologies have been developed to overcome these limitations. Alahakoon [1] designed a conversational AI system for cinnamon exports, which improved yield and price insights but was focused on large-scale operations with high computational requirements. Wijesekara [2] evaluated factors affecting cinnamon export income, achieving better pricing models, but its data-driven approach lacked integration with mobile tools for smallholders. Rahim [7] investigated machine learning for crop price prediction in Sri Lanka, preserving accuracy in vegetable markets but deemed too generalized for spices like cinnamon. Silva [9] developed an intelligent management system for Sri Lankan spices, enhancing operations but incurring high development costs. Perera [8] conducted a study on uplifting the Ceylon cinnamon industry with AgroX, targeting efficiency, and identified data access as a barrier for smallholder farmers. These studies collectively highlight those advanced methods improve accuracy but are not affordable or practical for small-scale cinnamon farmers.

The integration of AI/ML technologies in agriculture has shown promise for prediction and automation. Sharma [12] developed a cloud-enabled crop recommendation platform using ML, achieving improved precision in agriculture, demonstrating the feasibility of low-cost models for rural applications. Li [13] proposed data preprocessing techniques for agricultural ML models, emphasizing normalization and feature engineering, though their focus was on general crops. However, AI/ML applications for spice grading and pricing,

particularly cinnamon, remain underexplored, with no studies addressing the specific needs of smallholder cinnamon farmers [7], [12], [13].

Quality and economic considerations are crucial for cinnamon assessment. Fernando [5] studied the Ceylon cinnamon value chain, identifying optimal grading criteria to preserve value, but their approach lacked automation. Ranaweera [11] designed a cloud-based system for enhancing agricultural productivity, improving data handling for cinnamon, but it lacked real-time prediction features. Economic analyses by Wijesekara [2] and Alahakoon [1] highlight the high costs of advanced systems, underscoring the need for a cost-effective solution tailored to smallholder farmers.

The literature reveals several gaps: existing technologies are designed for large-scale operations, with high costs and complex setups [1], [5], [8]; AI/ML applications are limited to general crops or vegetables, not spices like cinnamon [7], [12]; and there is a lack of integrated, low-cost systems with mobile interfaces and predictive analytics for rural farmers [11], [13]. These gaps justify the need for a low-cost, AI/ML-based quality and price prediction tool to improve cinnamon assessment and accessibility for smallholder farmers in Sri Lanka.

1.2 Research Gap

The literature survey identifies the following critical gaps:

• Most quality and pricing tools (e.g., conversational AI systems, advanced market models) are costly and complex, making them unsuitable for smallholder cinnamon farmers [1], [5], [8].

• AI/ML-based quality and price prediction systems have been developed for general crops or vegetables but not for cinnamon, leaving a gap in spice-specific applications [7], [12].

• Current systems lack user-friendly mobile or web interfaces with multilingual support for real-time predictions, which are essential for rural adoption [11], [13].

• There is no low-cost, AI/ML-based tool that combines manual input-based quality grading and price prediction with accessible interfaces for cinnamon farmers [2], [10].

1.2.1 Research Gap and Novelty Analysis

To identify the research gap and highlight the novelty of the proposed AI/ML-Based Cinnamon Quality and Price Prediction Tool, a comparative analysis was conducted against selected existing works from the provided references ([1], [2], [5], [7], [10], [12]). These were chosen as representative studies covering advanced quality and pricing technologies for cinnamon ([1], [2], [5], [10]) and AI/ML applications in agriculture ([7], [12]). The comparison focuses on key features relevant to smallholder farmers in Sri Lanka, such as cost-effectiveness, AI/ML integration, manual input support, and farmer accessibility. The results are presented in Table 1, where "Yes" indicates the presence of a feature, "No" indicates its absence, and "Partial" indicates limited or indirect support.

The proposed tool stands out by integrating AI/ML-based quality grading and price prediction using manual inputs, offering a low-cost, IoT-independent solution with multilingual mobile/web interfaces—features not fully addressed in any single existing work. This addresses the gaps in affordability, spice-specific AI applications, and user-friendly prediction for rural farmers.

Feature	Research [1]	Research [2]	Research [3]	Research [7]	Research [10]	Research [12]	Proposed System
Specific to Cinnamon	Yes	Yes	Yes	No	Yes	No	Yes

Low-Cost for Smallholder Farmers	No	No	No	Partial (ML-based, but not for smallholders	No	Partial (cloud-based, but not spice-focused)	Yes
AI/ML Integration	Yes	Partial (data analysis)	No	Yes	No	Yes	Yes
Manual Input Support	No	No	No	No	No	No	Yes
Real-Time Prediction	Yes	Partial (export insights)	No	Yes	NO	Yes	Yes
Mobile/Web Interface	No	No	No	No	No	Partial (cloud access)	Yes
Multilingual Support	No	No	No	No	No	No	Yes
Techno-Economic Analysis for Small-Scale	Yes (but high cost highlighted)	Yes (field-tested)	No	No	No	No	Yes

Table 1: Comparative Analysis Highlighting the Novelty of the Proposed System

Explanation of Research Gap and Novelty

- **Research Gap:** Existing cinnamon-specific studies ([1], [2], [5], [10]) emphasize advanced quality and pricing methods (conversational AI, market models) that improve insights but lack manual input-based AI/ML integration and mobile accessibility, making them unsuitable for smallholder farmers due to high costs and complexity. AI/ML-focused works ([7], [12]) demonstrate prediction

capabilities but are limited to general crops or vegetables, without adaptation to cinnamon's unique grading needs (e.g., weight, moisture) or rural user-friendliness. None combine low-cost design with spice-specific, manual-input AI/ML features, leaving a gap in accessible, predictive solutions for small-scale cinnamon assessment in volatile markets like Sri Lanka.

- **Novelty of Proposed System:** The proposed tool uniquely addresses these gaps by offering an AI/ML framework tailored for cinnamon, incorporating low-cost models (e.g., Random Forest, XGBoost inspired by [7], [12]), manual input support for quality and price prediction (building on [4]'s data-driven insights), real-time predictions with multilingual interfaces, and a focus on economic empowerment. This makes it the first farmer-friendly, IoT-independent solution for smallholders, enhancing adoption and market competitiveness while reducing reliance on middlemen noted in [2] and [3].

1.3 Research Problem

Smallholder cinnamon farmers in Sri Lanka rely on manual grading and pricing methods, which result in inconsistent quality assessments, reduced income, and limited market competitiveness. Advanced AI/ML technologies exist but are too costly and complex for rural adoption. Therefore, the research problem addressed in this study is:

"How can a low-cost, AI/ML-based quality and price prediction tool be developed to provide automated grading and pricing for cinnamon bark using manual inputs, with real-time predictions and accessible mobile/web interfaces for smallholder farmers in Sri Lanka?"

2. OBJECTIVES

2.1 Main Objective

Provide farmers with cutting-edge tools for better quality, price, and industry decision-making by building a comprehensive AI/ML-based platform to increase Ceylon cinnamon production and export efficiency in Sri Lanka.

2.2. Specific Objectives

The following are the sub-objectives of conducting this research.

- Provide a real-time prediction solution accessible via mobile or web platforms.
- Tailor features to the Sri Lankan cinnamon farming context, including local language support.
- Improve farmers' decision-making and profitability.
- Enable early intervention through timely quality and price insights.
- Increase farmer engagement with data-driven agricultural practices.

Ensure a non-invasive, manual-input-based system requiring no costly hardware.

3. METHODOLOGY

3.1 System Overview

The CinnamonSri system is a comprehensive, integrated platform that uses the Internet of Things (IoT), machine learning (ML), artificial intelligence (AI), and user-friendly mobile applications to transform Ceylon cinnamon production in Sri Lanka. This solution's four interrelated components help smallholder farmers increase productivity, quality, and profitability while addressing major issues in the cinnamon sector, including uneven drying, manual grading, price volatility, and disease control.

1. Smart Drying Automation (IoT): This part uses DHT22 sensors to track the humidity and temperature in drying chambers in real time. An ESP8266 microcontroller regulates fans and heaters via relays to maintain optimal conditions. Data is sent to Firebase for storage and display via a mobile application that offers voice notifications in Sinhala. This minimizes deterioration from over- or under-drying and guarantees consistent export-grade quality and remote monitoring.

2. AI-Powered Quality & Price Prediction: Farmers use a mobile or online app to enter characteristics like as temperature, moisture content, bark thickness, and pre- and post-drying weights. ML models predict quality ratings (Premium, Grade A, and Grade B) and anticipated costs (LKR/kg). Examples of these models are Random Forest for classification and XGBoost for regression. This reduces dependency on middlemen and gives farmers data-driven insights to negotiate better prices.

3. A Guide to AI Cinnamon Harvesting: This AI-powered module uses trained models on cinnamon growth stage datasets to assess uploaded plant photos. By using regional planting calendars to determine maturity and health condition, it sends out time-based notifications. Alerts guide farmers on planting, care, and harvesting, optimizing yield and reducing losses from premature harvesting.

4. Disease Detection Module: Using the app, farmers can take or submit pictures of leaves or bark. Leaf spot is classified by a Convolutional Neural Network (CNN) model that was trained on datasets related to cinnamon sickness. Treatment recommendations are included in the results, along with voice notifications in Tamil and Sinhala for prompt action. Early intervention is made possible by the logging of data to a central dashboard for long-term monitoring, particularly during wet seasons.

These elements work together seamlessly through a cloud backend (like Firebase) and centralized mobile app, enabling data exchange and multilingual user interfaces. CinnamonSri fills technical gaps by integrating automation, predictive analytics, and localized alerts. This promotes sustainable practices and boosts financial returns for Sri Lanka's cinnamon industry. Figure 1 illustrates the CinnamonSri platform and its four interrelated components clearly.

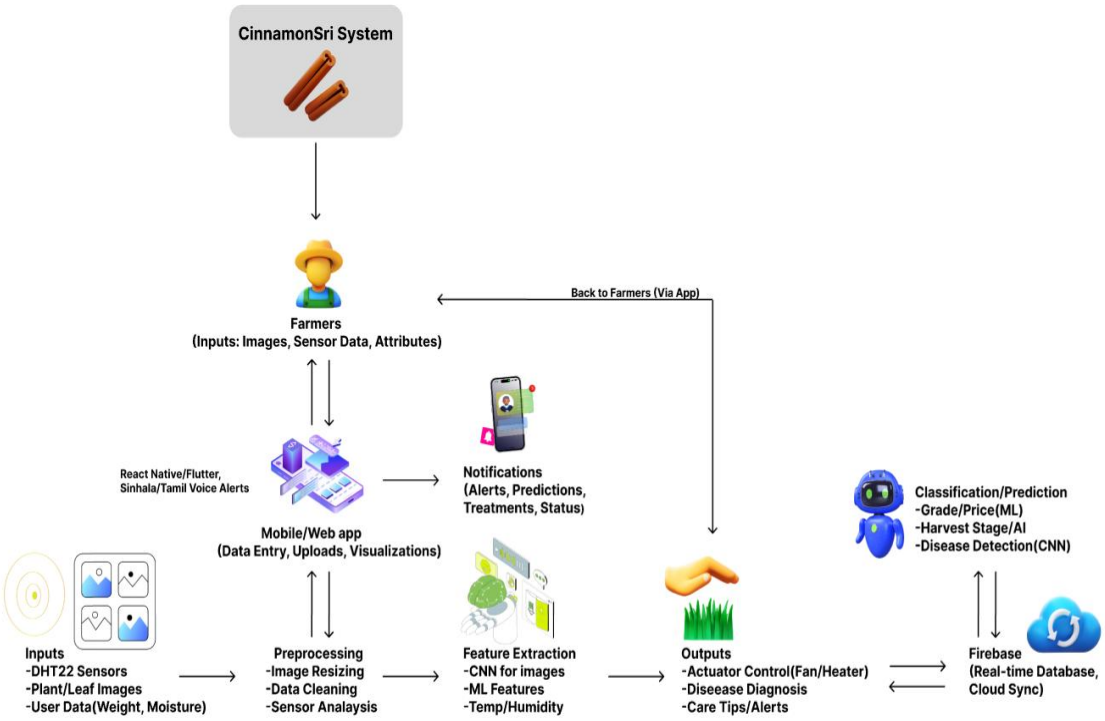

Figure 1: System Overview Architecture

3.2 Individual Component Overview

By offering precise evaluations of cinnamon quality and market pricing, the CinnamonSri system's AI-Powered Quality & Price Prediction component aims to improve decision-making for Sri Lankan cinnamon growers. Through an intuitive mobile or web application, this module enables farmers to manually enter important information, including temperature, moisture content, bark thickness, and pre- and post-drying weights. The system processes these inputs to provide real-time predictions by utilizing machine learning models, such as Random Forest for identifying quality grades (e.g., Premium, Grade A, Grade B) and XGBoost for regressing anticipated pricing (LKR/kg). These forecasts lessen financial uncertainty by enabling farmers to improve their selling tactics and bargain with middlemen for lower pricing.

The component uses a cloud-based backend to store and analyze data, and it connects easily with the larger CinnamonSri platform. The interface offers clear outputs like quality grades and price projections, and it supports both Tamil and Sinhala, making it accessible to local farmers. This method provides a cost-effective solution that is suited to the needs of smallholders by reducing dependency on expensive gear. The lesson tackles the issue of inconsistent pricing and grading by offering practical insights, which helps the Ceylon cinnamon business become more profitable and export competitive.

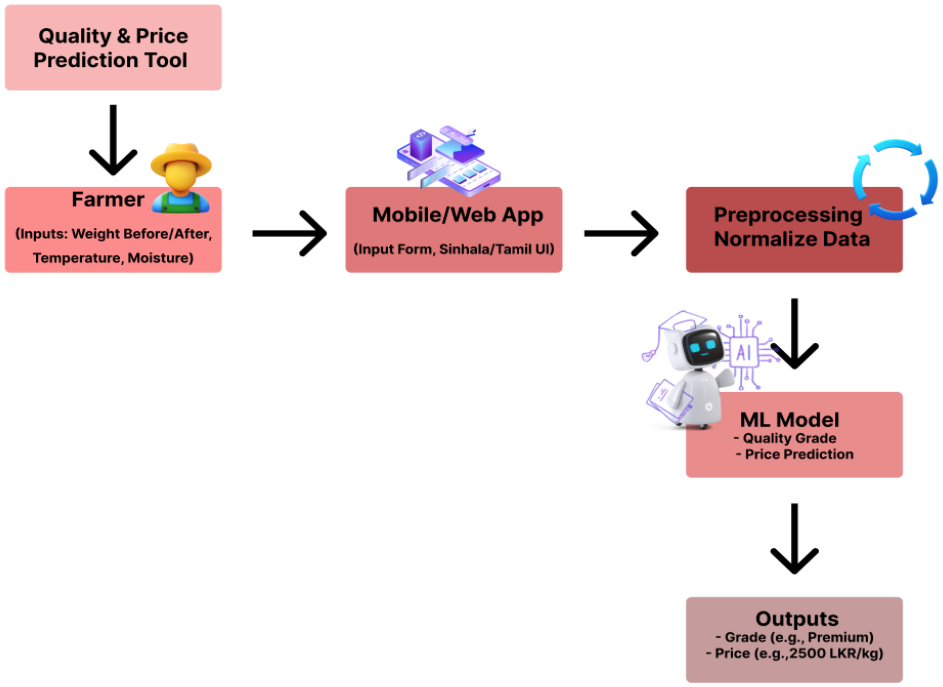

Figure 2: Individual Component Overview Architecture

3.3 Requirement Gathering

Requirement gathering was conducted through an in-depth review of recent research on AI/ML applications in agriculture, analysis of existing cinnamon production systems, and exploration of online resources related to quality grading and pricing. Additionally, real-world data from Sri Lankan cinnamon farmers, collection centers, and agricultural institutions were examined to identify key challenges and needs, ensuring the development of a practical and relevant solution.

3.4 Past Research Analysis

In the context of past research analysis, numerous studies and publications focus on AI and machine learning applications in agriculture, with a growing interest in quality grading and price prediction. However, fewer publications specifically address cinnamon

production, particularly without IoT dependency. Key topics of interest include Random Forest, XGBoost, and data preprocessing techniques.

The analysis concentrated on identifying the methodologies and tools employed in existing agricultural prediction models, such as those using scikit-learn and TensorFlow. It also highlighted challenges faced by previous researchers, including limited localized datasets, lack of cost-effective solutions, and difficulties in validating predictions across diverse farming contexts, guiding the development of a tailored approach for cinnamon farmers.

3.5 Feasibility Study

3.5.1 Technical Feasibility

3.5.1.1 Knowledge on Technologies

Understanding of the following technologies should be necessary in order to create the suggested AI/ML-Based Cinnamon Quality and Price Prediction Tool:

1. **Random Forest:** Random Forest provides reliable performance with small datasets and is used to categorize cinnamon quality grades (e.g., Premium, Grade A, Grade B) based on input parameters like weight variations and moisture content.
2. **XGBoost:** XGBoost provides excellent accuracy and efficiency for predictive modeling by regressing estimated pricing (LKR/kg) by assessing parameters like temperature and bark thickness.
3. **Data Preprocessing:** Using libraries like Pandas and NumPy, methods like normalization and encoding are crucial for converting unprocessed farmer inputs into a format that is appropriate for machine learning models.
4. **scikit-learn:** This package provides training, validation, and evaluation methods to guarantee model reliability and is used to create Random Forest and XGBoost models.

5. Flask and Flutter: Flask is used to create web-based user interfaces, and Flutter facilitates the creation of mobile applications, allowing farmers to access multilingual interfaces.

3.5.1.2 Knowledge on Tools

In order to develop the proposed AI/ML-Based Cinnamon Quality and Price Prediction Tool, all members should have a thorough understanding of development tools, project management, and supportive software. The figure illustrates the current project management board using a tool like MS Project or Trello, outlining tasks, timelines, and responsibilities for the development process.

3.5.1.3 Data collection Knowledge

Data collection and preprocessing are critical steps in building the prediction model.

1. **Data Collection:** Methods include interviewing cinnamon farmers, conducting surveys with collection centers, and collaborating with agricultural institutions to gather relevant data such as weights, moisture levels, and market prices.
2. **Data Preprocessing:** This involves data cleaning to handle missing values, data transformation to normalize inputs, data reduction to focus on key features, and data integration to combine diverse sources using tools like Pandas and NumPy.

3.5.2 Schedule Feasibility

3.5.3 Economic Feasibility

The CinnamonSri system's AI-Powered Quality & Price Prediction component is made to be financially feasible for Sri Lankan cinnamon growers, especially smallholders who are the foundation of the sector. By utilizing current mobile technologies and human data entry, this module does not require costly IoT gear, thus lowering the initial investment

expenses. Development uses cloud services like Firebase and open-source technologies like scikit-learn and XGBoost to ensure scalability and save operating costs.

The financial gain comes from providing farmers with precise quality ratings and pricing forecasts, which might boost their income by 20% to 30% by enabling them to make well-informed selling decisions and lessen their reliance on intermediaries. Long-term sustainability is ensured by the system's inexpensive maintenance, which is bolstered by recurring model updates utilizing locally gathered data. Improved export quality and market competitiveness can offset the low initial startup expenses, which mostly include app development and farmer training. Within the first year of implementation, this feasibility analysis predicts a positive return on investment.

3.6 System Analysis

3.6.1 Software Solution Approach

Below mentioned are the approach of the solution development, and Figure illustrates the flow of the solution for the AI/ML-Based Cinnamon Quality and Price Prediction Tool.

1.Data Collection: Gather data through interviews with cinnamon farmers, surveys from collection centers, and collaboration with agricultural institutions to obtain inputs like weight before/after drying, temperature, moisture content, and bark thickness.

2. Data Input: Farmers manually enter processed data into a mobile or web app, ensuring accessibility without IoT hardware.

3. Preprocessing: Perform preprocessing tasks such as normalizing numerical data (e.g., weight, moisture) to a 0–1 scale, handling missing values with averages, and encoding quality grades (e.g., Premium = [1,0,0]) using Python libraries like Pandas and NumPy.

4. Feature Extraction: Extract meaningful features from the preprocessed data, focusing on weight differences, temperature, and moisture levels, which are critical for quality and price prediction.

5. Model Training: Train classification models (Random Forest) for quality grades and regression models (XGBoost) for price prediction, utilizing an 80/20 train-test split and cross-validation for optimization.

6. Model Evaluation: Evaluate model performance using metrics such as F1-score for classification (>85% anticipated) and Mean Squared Error for regression (<10% variance), ensuring reliability.

7. Prediction Deployment: Deploy the trained models via a cloud-based API (e.g., AWS) or local device, integrating with the app to provide real-time quality grades and price estimates to farmers.

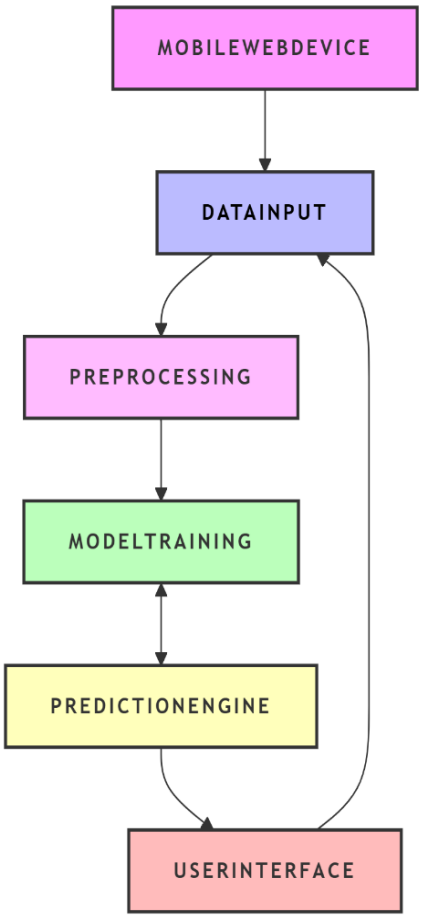

Figure 3: General Flow of Cinnamon Quality and Price Prediction

3.6.2 Tools & Technology

- Programming Languages: Python
- Frameworks: scikit-learn, XGBoost, TensorFlow (for potential deep learning enhancements), Flask (web app), Flutter (mobile app)
- Database Systems: Firebase (for storing farmer input data and predictions)
- Web Development Frameworks
  - React JS: Front-end development for the web interface
  - Tailwind CSS: UI component design and development
- UI Design Implementation: Figma
- Diagramming Too: Draw.io
- Integrated Development Environment (IDE): VS Code
- Version Control System (VCS): Git (GitHub)
- Tools: Microsoft Teams

- Testing Tools
  - Pytest: Unit testing for Python
  - Postman: API testing
- Deployment Tools
  - Docker: Containerization
  - Kubernetes: Orchestration
  - Jenkin: Automation
  - Ansible: Configuration
- Code Quality Assurance Too: SonarQube
- Project Management: MS Project

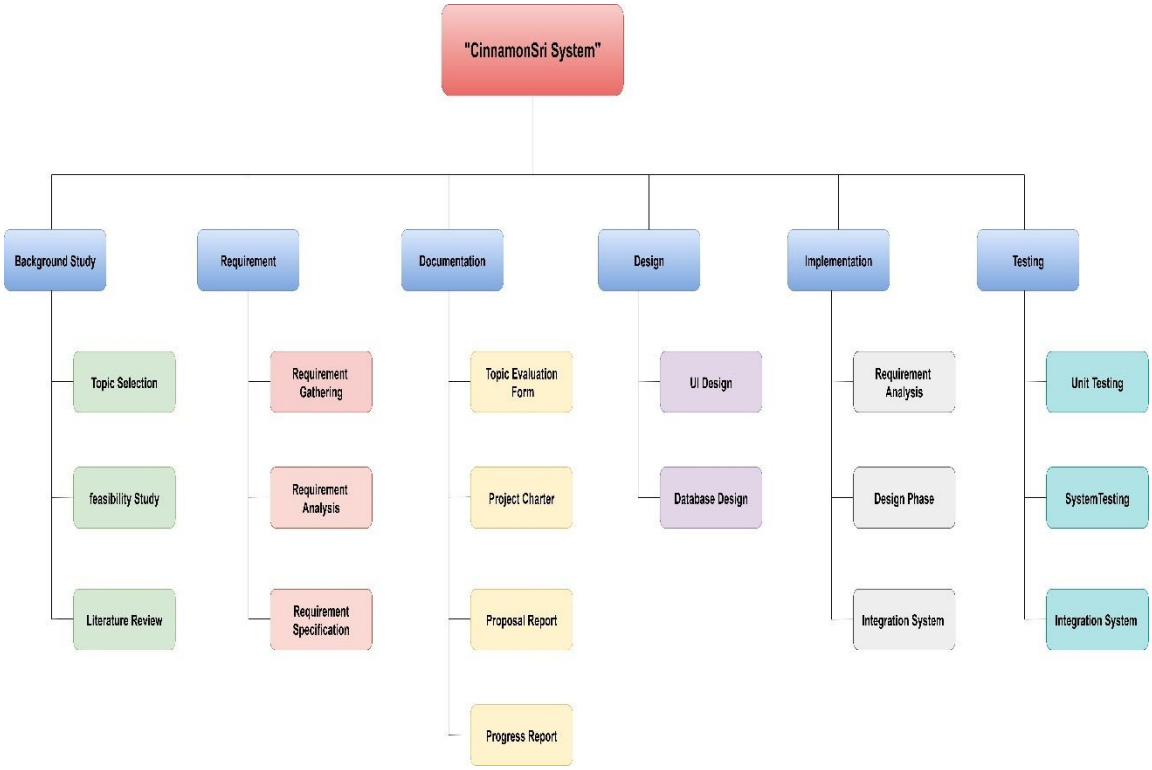

Figure 4: Work breakdown Structure

3.7 Project Requirements

3.7.1 Functional Requirements

The functional requirements for the proposed AI/ML-Based Cinnamon Quality and Price Prediction Tool are as follows:

- Collect data specific to the Sri Lankan cinnamon farming context.
- Predict cinnamon quality grades (e.g., Premium, Grade A, Grade B) and estimated prices (LKR/kg) based on farmer inputs.
- Provide real-time predictions to farmers through a mobile or web app interface. Collected the data for Sri Lankan context.

3.7.2 Non-Functional Requirements

The non-functional needs that were prioritized throughout the creation of the suggested tool are listed below:

- Usability: Ensure an intuitive interface accessible to farmers with minimal technical skills.
- Reliability: Guarantee consistent and accurate predictions under varying input conditions.
- Availability: Make the tool accessible 24/7 via cloud or local deployment.
- Accuracy: Achieve high prediction accuracy (>85% for quality, <10% error for price).
- Performance: Ensure quick processing and response times for real-time use. Usability

3.8 Testing

Testing plays a crucial role in the success of the AI/ML-Based Cinnamon Quality and Price Prediction Tool. To evaluate the functionality of the proposed solution, the initial plan involves testing among team members and peers, serving as unit or system testing to identify early issues. This will include validating input processing and model outputs with sample data.

In the final phase of the research, the tool will be tested with a selected group of Sri Lankan cinnamon farmers in both local and deployed environments, functioning as acceptance testing (Alpha and Beta testing). This will assess real-world usability and prediction accuracy. Testing will primarily rely on manual methods, with occasional use of internal functions and assert statements in Python to verify model performance and data integrity.

3.8.1 Test Cases

Tentative test cases ensure the system meets functional and non-functional requirements. These will be refined during development.

Test Case ID	Test Case Name	Objective	Procedure	Expected Result
TC-001	Input Validation	Verify that the system accurately processes manual input data.	Enter sample inputs (e.g., Weight Before = 1000g, Weight After = 850g, Temperature = 40°C, Moisture = 12%) and check for errors or invalid data handling.	System accepts valid inputs and flags invalid entries (e.g., negative weights) with an error message within 5 seconds.
TC-002	Prediction Accuracy	Ensure the AI/ML models provide accurate quality grades and price predictions.	Input a dataset of 50 known samples (e.g., Weight Before = 1000g, Weight After = 850g, Temperature = 40°C, Moisture = 12%) and compare predicted grades (Premium) and prices (2500 LKR/kg) against actual values.	Accuracy exceeds 85% for quality classification (F1-score) and Mean Squared Error is below 10% for price regression.
TC-003	Mobile/Web Interface Response	Confirm that predictions are delivered in real-time via the interface.	Input data (e.g., Weight Before = 950g, Weight After = 800g) and measure the time to display results on the mobile/web app.	Results (e.g., Grade A, 2000 LKR/kg) displayed within 10 seconds.
TC-004	Multilingual Usability	Verify that the interface is user-friendly with Sinhala/Tamil	Conduct usability testing with 10 farmers, assessing ease of navigation and language	80% of users complete tasks (e.g., input data, view predictions) without assistance.

		support for rural farmers.	comprehension in Sinhala and Tamil.	
TC-005	Cloud Integration Performance	Ensure the cloud-based backend handles data storage and prediction requests efficiently.	Simulate 100 concurrent user inputs over 1 hour and monitor response time and data integrity.	Response time remains below 15 seconds, with no data loss or corruption.

Table 2: Test Cases

3.9 Timeline

The proposed timeline for the project is as follows.

Figure 5: Gantt Chart for Project Schedule Management

4.0 BUDGET

Since the outcome of the proposed AI-Powered Quality & Price Prediction component is a software-based solution, there are no hardware components associated with the implementation. The primary cost will stem from subscription fees to cloud providers for computing power and data storage (e.g., Firebase or AWS). However, additional costs are anticipated as outlined in the table below.

Type	**Cost**
Internet use and app hosting	6000 LKR
Publication costs	15000 LKR
Training materials	7000 LKR
TOTAL	28000 LKR

Table 3: Cost Management Plan

This amount may differ according to the economic crisis.

5.0 SUMMARY

A novel research area, "AI-Powered Quality & Price Prediction for Agriculture," focuses on utilizing artificial intelligence and machine learning to assess cinnamon quality and predict market prices based on farmer inputs. This study employs advanced ML algorithms, such as Random Forest and XGBoost, to analyze parameters like weight, moisture content, and drying conditions, providing actionable insights for farmers. The goal is to enhance decision-making, optimize pricing strategies, and address challenges in traditional cinnamon production.

This research holds significant potential for the agricultural sector, particularly in improving export competitiveness and profitability for Sri Lankan cinnamon farmers. It can empower smallholders with data-driven tools, reduce reliance on middlemen, and support sustainable practices. By integrating this solution into a mobile/web platform, the study offers a scalable approach to benefit farmers, cooperatives, and export stakeholders, fostering economic resilience and quality enhancement in the cinnamon industry.

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